

Instructions

There are 12 questions on this examination paper. Answer **all** questions.

Questions do not necessarily carry equal marks. To help you manage your time during this examination, a maximum time for each question is suggested. If you remain within these times you should have about 10 minutes left to review your work.

Write your answers in blue or black pen. You may use pencil in graphs and diagrams only.

This examination booklet will be scanned and your work will be presented to an examiner on screen. Anything that you write outside of the answer areas may not be seen by the examiner.

Write your answers into this booklet. There is space for extra work at the back of the booklet. Label any extra work clearly with the question number and part.

The superintendent will give you a copy of the *Formulae and Tables* booklet. You must return it at the end of the examination. You are not allowed to bring your own copy into the examination.

You may lose marks if your solutions do not include supporting work.

You may lose marks if you do not include the appropriate units of measurement, where relevant.

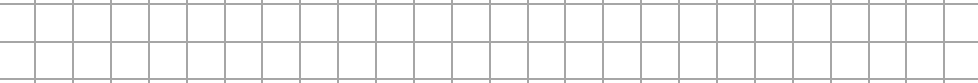
You may lose marks if you do not give your answers in simplest form, where relevant.

Write the make and model of your calculator(s) here:

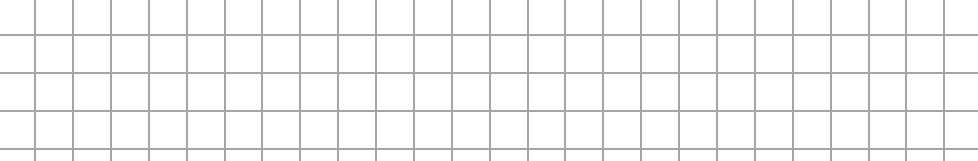
Question 1

(Suggested maximum time: 10 minutes)

- (a)** An airline allows carry-on luggage with a maximum volume of $52\,800\text{ cm}^3$.
A suitcase in the shape of a cuboid has length 40 cm and width 24 cm . What is the height of the suitcase if it is the maximum volume allowed for carry-on luggage?

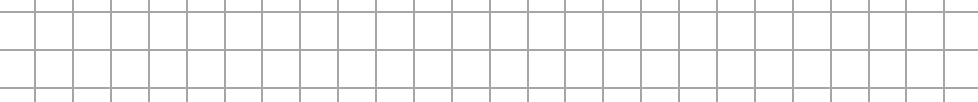


- (b)** A flight from Dublin to New York travelled a distance of 5112 km at an average speed of 818 km per hour.
How long in **hours and minutes** did the flight take?
Give your answer correct to the nearest minute.



- (c) A passenger, on the return flight, converts \$750 US dollars to euro. The exchange rate is €1 = \$1.06.

Work out **in euro** how much the passenger receives.
Give your answer correct to the nearest cent.



Question 2 (Suggested maximum time: 5 minutes)

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- (a) A sports club runs a weekly lotto draw where a person can choose any 4 numbers from 1 to 24 inclusive. A person can win or share in the jackpot if all of their chosen numbers are drawn.

Nila picks 4 numbers and enters them into the weekly lotto draw.

She tells her friend “I have a 50:50 chance of having one of my numbers drawn first”.

Put a tick (✓) in the correct box to show which statement is true.

Tick **one** box only. Give a reason for your answer.

Nila's statement is correct

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Nila's statement is incorrect

7

Reason:

- (b)** A particular coin was tossed several times.

The relative frequency of this coin landing on tails was found to be 0.3.

- (i) What is the relative frequency of this coin landing on heads?

[illegible]

- (ii) This coin landed on tails 150 times.

Work out the total number of times that the coin was tossed.

[illegible]

- (iii)** In your opinion, is the coin biased or not biased?

Put a tick (✓) in a box below to indicate your answer.

Tick **one** box only. Give a reason for your answer.

The coin is biased

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The coin is not biased

7

Reason:

Question 3 (Suggested maximum time: 5 minutes)

Question 3 (Suggested maximum time: 5 minutes)

- (a)** Factorise $x^2 - 121$.

[illegible]

- (b)** Factorise fully $y^2 - 8cp + 8y - cpy$.



- (c) (i) The sets A and B are as follows:

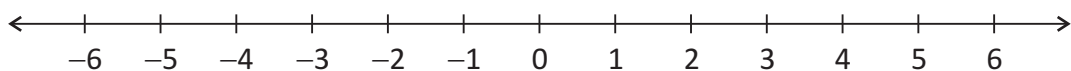
A is the set of all numbers x such that $3x - 5 < 13$ where $x \in \mathbb{Z}$.

B is the set of all numbers x such that $3x - 5 \geq -11$ where $x \in \mathbb{Z}$.

Use algebra to solve the inequalities and hence write down the solution sets of A and B .

$3x - 5 < 13$	$3x - 5 \geq -11$
$A =$	$B =$

- (ii) Show the elements of $A \cap B$ on the number line below.



Question 4

(Suggested maximum time: 15 minutes)

Aoife buys a new car.

The car comes in 5 different colours (Black, Red, Blue, Silver and White), 4 different body shapes (Saloon, Hatchback, Coupe and Convertible) and 3 different engine types (Electric, Hybrid and Petrol).



- (a)** Work out how many options Aoife has when choosing her new car.

[illegible]

- (b)** Aoife considers two finance options to buy the car.
The terms of these options are shown in the table below.

Credit Union Loan	Car Dealership Finance
Repayments over 4 years of €748 per month.	Repayments of 35 instalments of €880 per month and a final payment of €5000 due at the end of the instalment phase.

- (i)** Which finance option is the more cost-effective for Aoife?
Give a reason for your answer.

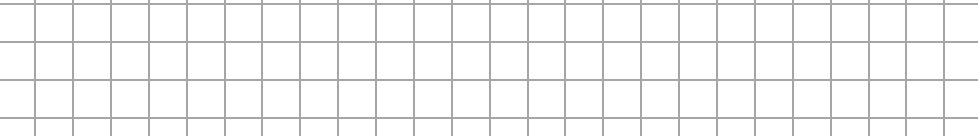
Answer:	
Reason:	

- (ii) Give a reason why Aoife might choose the **less** cost-effective option.

Reason:



- (c) Aoife's new car costs €32 200. Seán says her car will decrease in value by 15% in the first year. If Seán is correct, work out the value of Aoife's car after 1 year.



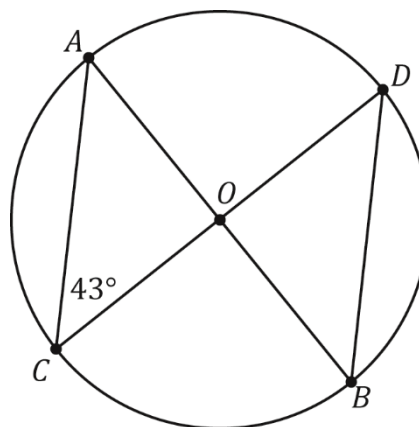
- (d)** Simon bought the same model of car as Aoife for the same price. Simon saved €10 000 at the start of each year for 3 years in a savings account that offered an interest rate of 3.5% per year compounded annually. If Simon used all his savings to buy his car, work out how much more he needed to complete the purchase. Give your answer correct to the nearest euro.

[illegible]

Question 5

(Suggested maximum time: 5 minutes)

- (a) The diagram shows a circle (not to scale).
 $[AB]$ and $[CD]$ are diameters of the circle and intersect at the point O .
 $|\angle ACD| = 43^\circ$.

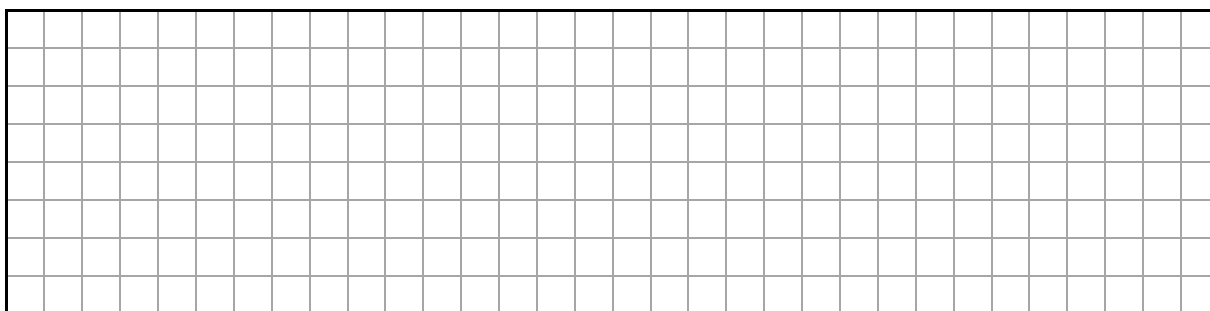


- (i) Write down the size of the angle $\angle ABD$ and the size of the angle $\angle BDC$.

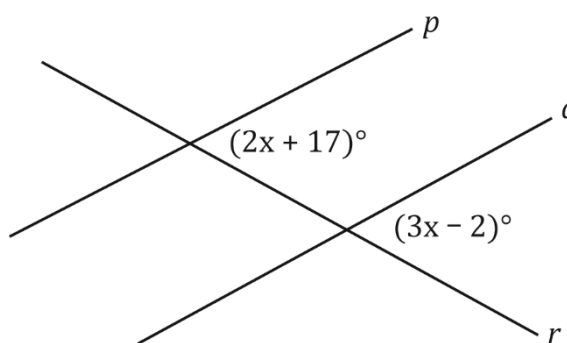
$|\angle ABD| = \boxed{}$

$|\angle BDC| = \boxed{}$

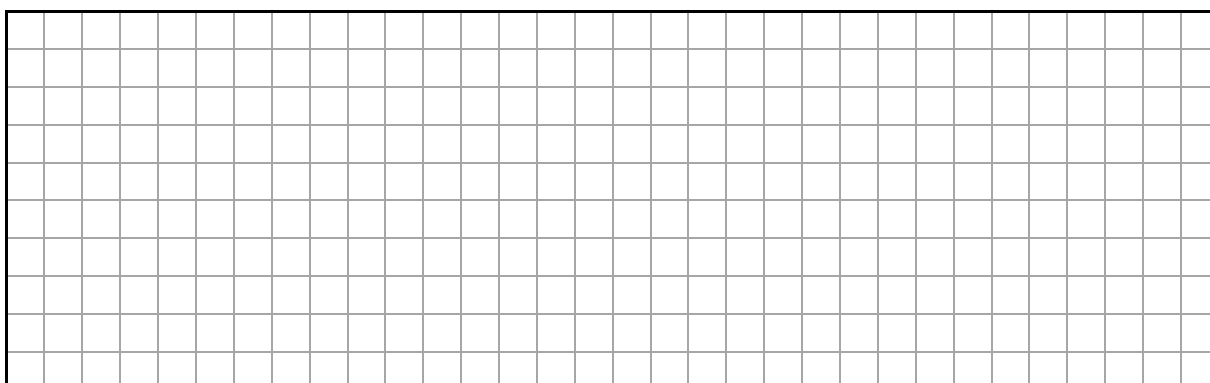
- (ii) Work out the size of the angle $\angle COB$.



- (b) The diagram shows the lines p , q and r .
 p is parallel to q , and the angles
 $(2x + 17)^\circ$ and $(3x - 2)^\circ$ are shown.



Work out the value of x .



Question 6**(Suggested maximum time: 10 minutes)**

- (a)** Solve the equation

$$\frac{1-5x}{3} + \frac{1-3x}{2} = 4.$$

- (b)** Solve the following simultaneous equations to find the value of x and the value of y .

$$7x - 3y = 11.$$

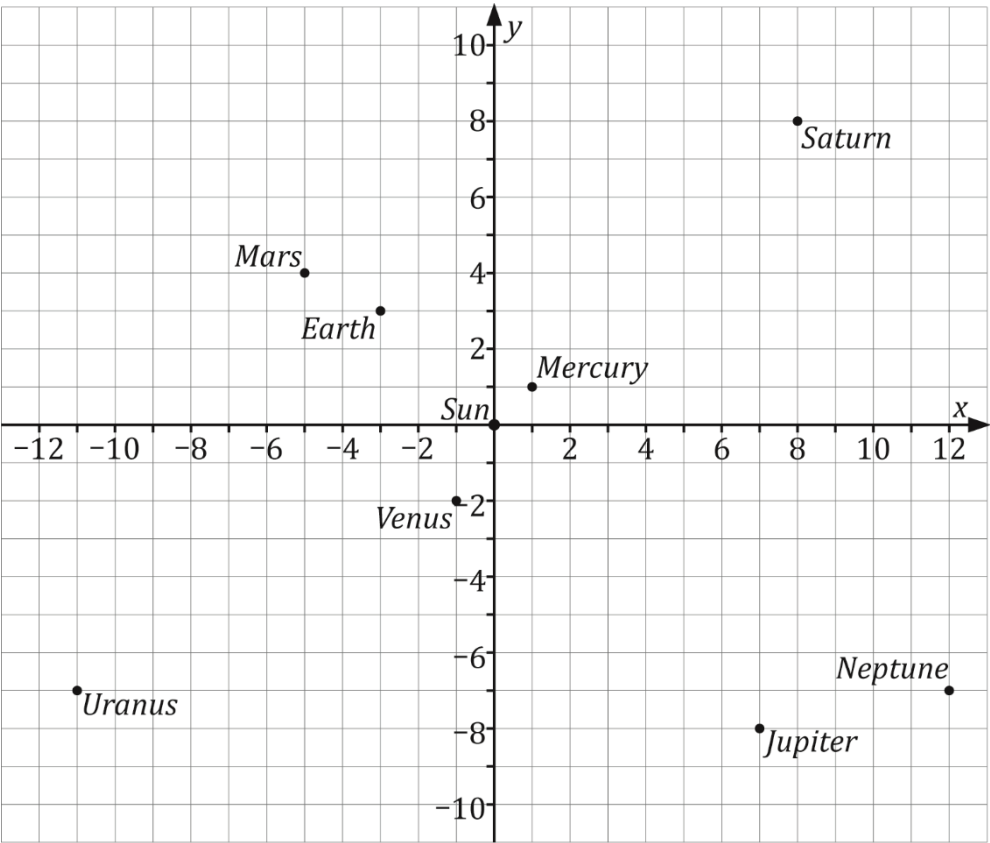
$$2x + 5y = 9.$$

- (c)** Verify your answers to part **(b)**.

Question 7

(Suggested maximum time: 15 minutes)

The co-ordinate diagram below shows the location of the Sun and the planets in our solar system.



- (a) The Sun has the co-ordinates (0, 0).
Write down the co-ordinates of the planets in the table below.

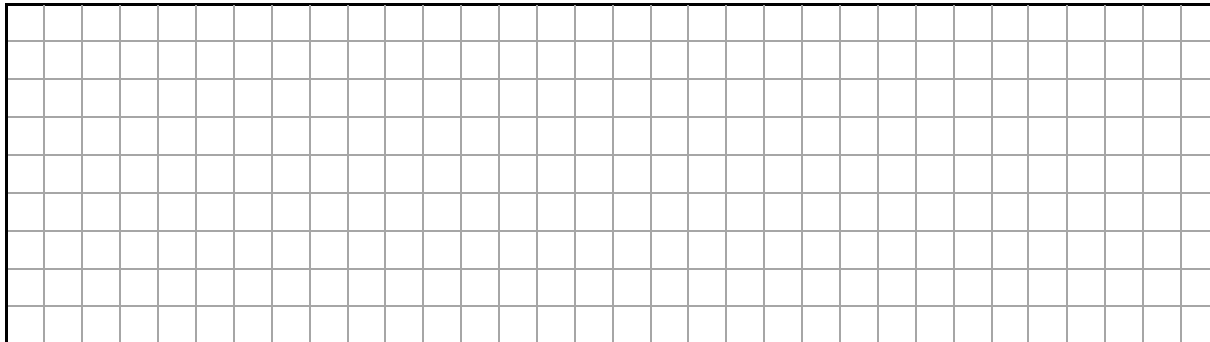
Planet	Mars	Earth	Venus	Uranus
Co-ordinates	(-5, 4)	(,)	(-1, -2)	(,)

Planet	Saturn	Mercury	Neptune	Jupiter
Co-ordinates	(8, 8)	(,)	(12, -7)	(,)

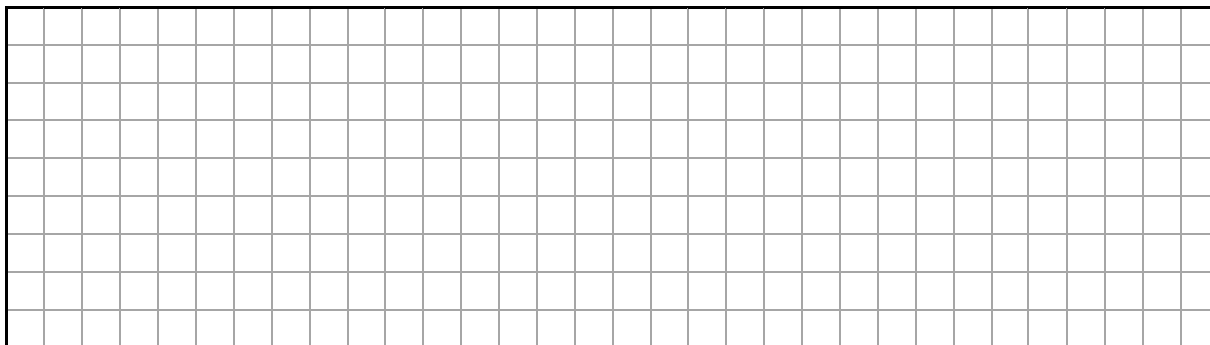
- (b) Work out the slope of the line that joins Mercury and Venus.

An asteroid, having entered our solar system, is currently located at co-ordinates $(-8, 2)$.

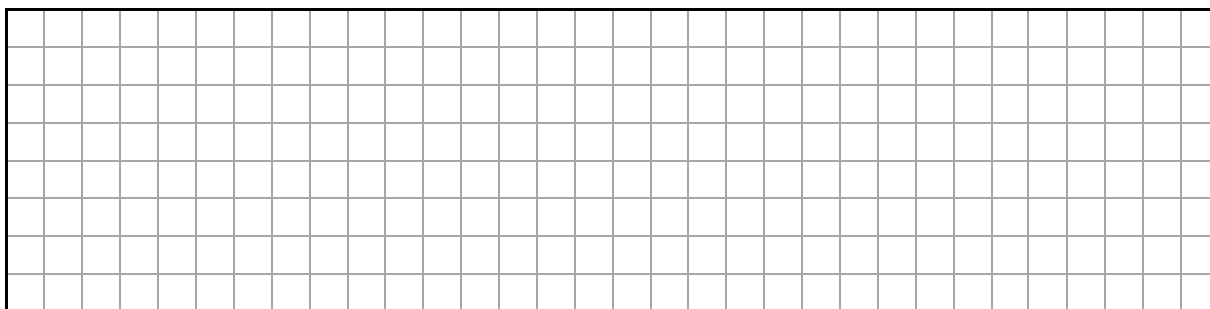
- (c) Plot the location of this asteroid on the co-ordinate grid above and label this point A .
- (d) The path of the asteroid is a straight line segment which is **perpendicular** to the line joining Mercury and Venus. Work out the equation of the path of the asteroid.
Give your answer in the form $ax + by + c = 0$, where $a, b, c \in \mathbb{Z}$.



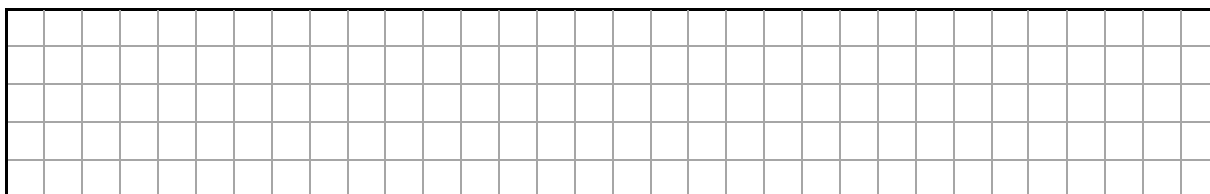
- (e) Investigate whether Jupiter is in the path of the asteroid.



- (f) NASA launched the Cassini space probe from Earth. It is currently halfway to Saturn. Find the current co-ordinates of the Cassini space probe on the co-ordinate grid.



- (g) Scientists say that Earth is 4,540,000,000 years old. Write the age of Earth in the form $a \times 10^n$, where $1 \leq a < 10$ and $n \in \mathbb{N}$.



Question 8

(Suggested maximum time: 10 minutes)

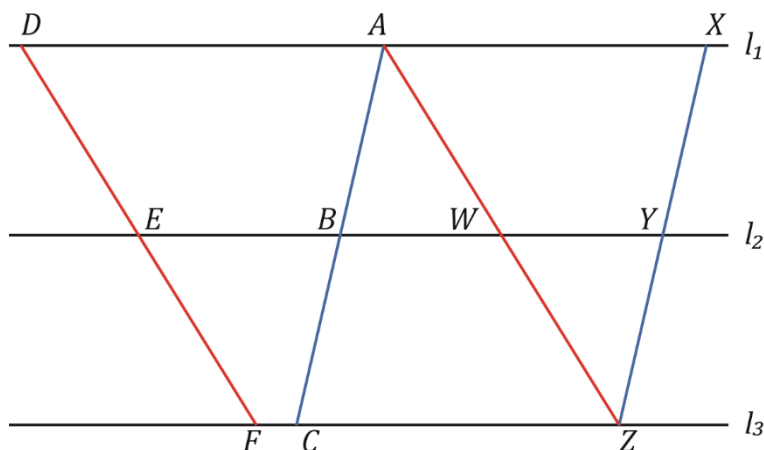
The geometric proof constructed below is as follows:

"If three parallel lines cut off equal segments on some transversal line, then they will cut off equal segments on any other transversal line."

The diagram shows three parallel lines, l_1 , l_2 and l_3 .
 AC and DF are transversal lines.
 $|AB| = |BC|$.

AZ is drawn parallel to DF ,
cutting l_2 at W .

XZ is drawn parallel to AC ,
cutting l_2 at Y .



Fill in the missing reasons in the table below to **prove that $|DE| = |EF|$** .

Statement	Reason
$ YZ = BC $	
$\Rightarrow YZ = AB $	$ AB = BC $
$ \angle BAW = \angle WZY $	
$ \angle AWB = \angle ZWY $	
$\Rightarrow \angle ABW = \angle ZYW $	Third angle in the triangle
$\therefore \triangle ABW$ and $\triangle WZY$ are congruent.	
$\therefore AW = WZ $	Corresponding sides in congruent triangles
But $ AW = DE $ and $ WZ = EF $	
$\therefore DE = EF $	QED

(Suggested maximum time: 10 minutes)

The graph shows a journey starting at 10:00. The distance from home increases linearly to 90 km at 11:45. From 11:45 to 12:45, the distance remains constant at 90 km.

Time	Distance from home (km)
10:00	0
11:45	90
12:45	90

-

- _____

- [illegible]

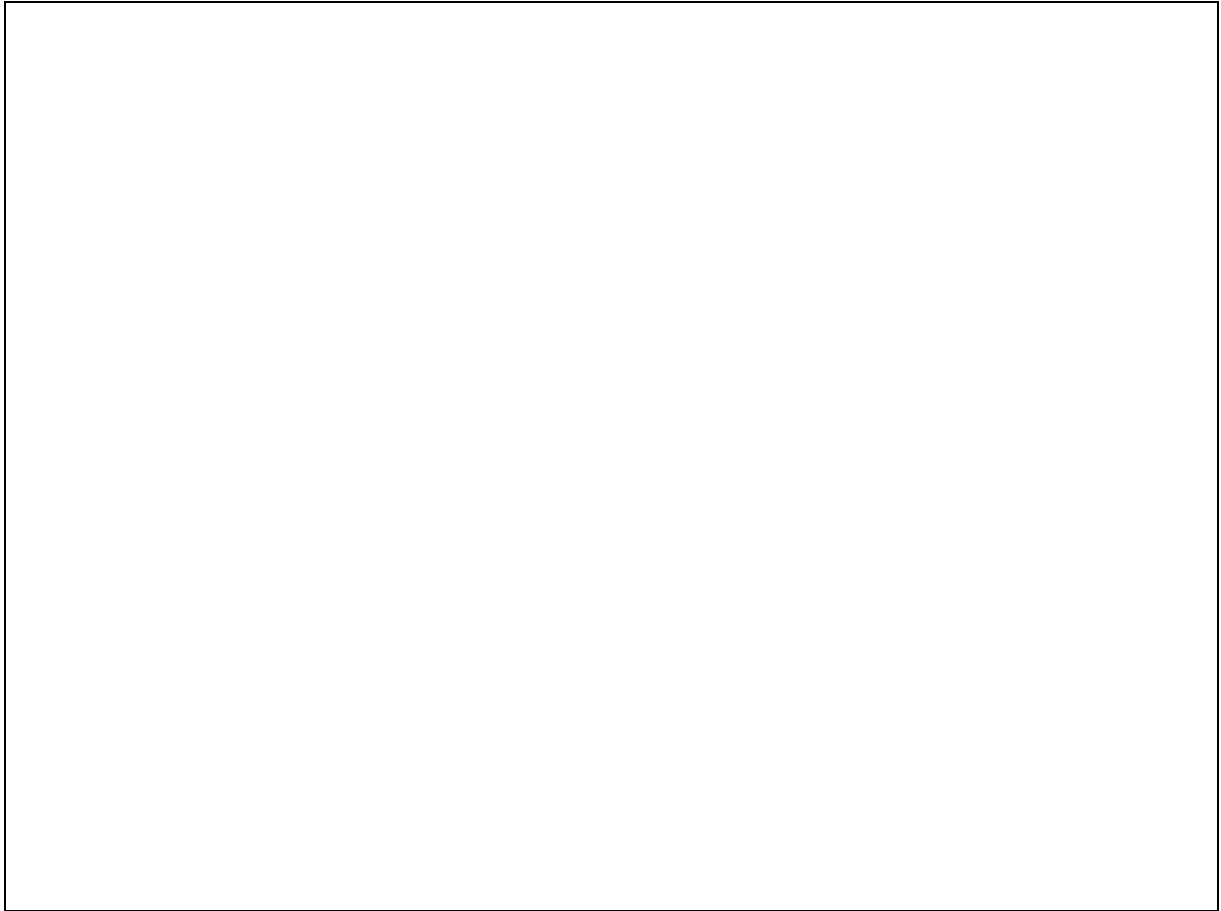
- km

- hours

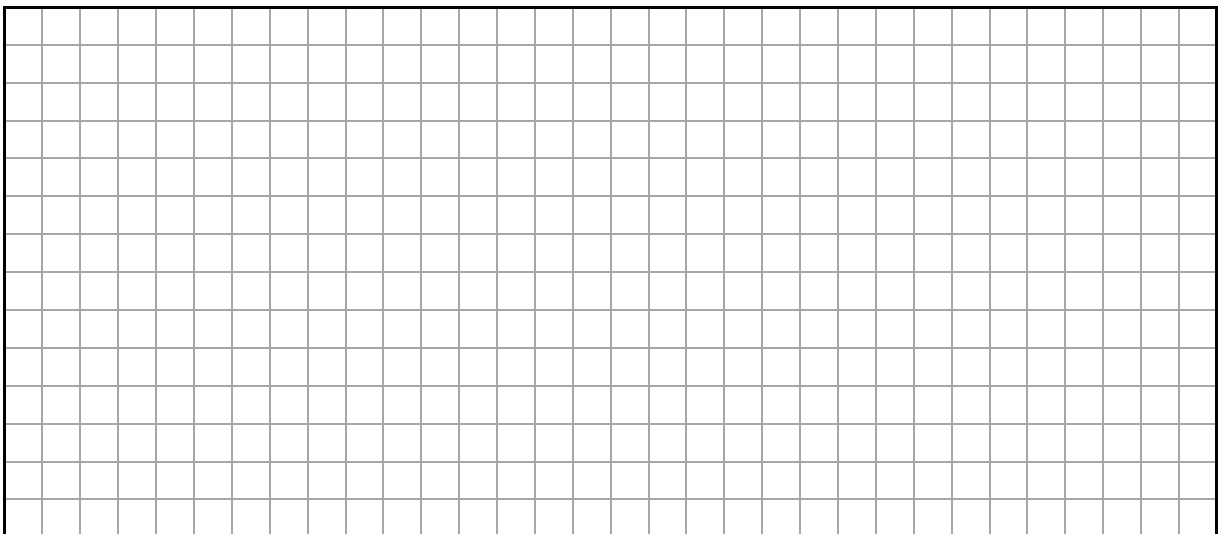
[illegible]

Question 10**(Suggested maximum time: 10 minutes)**

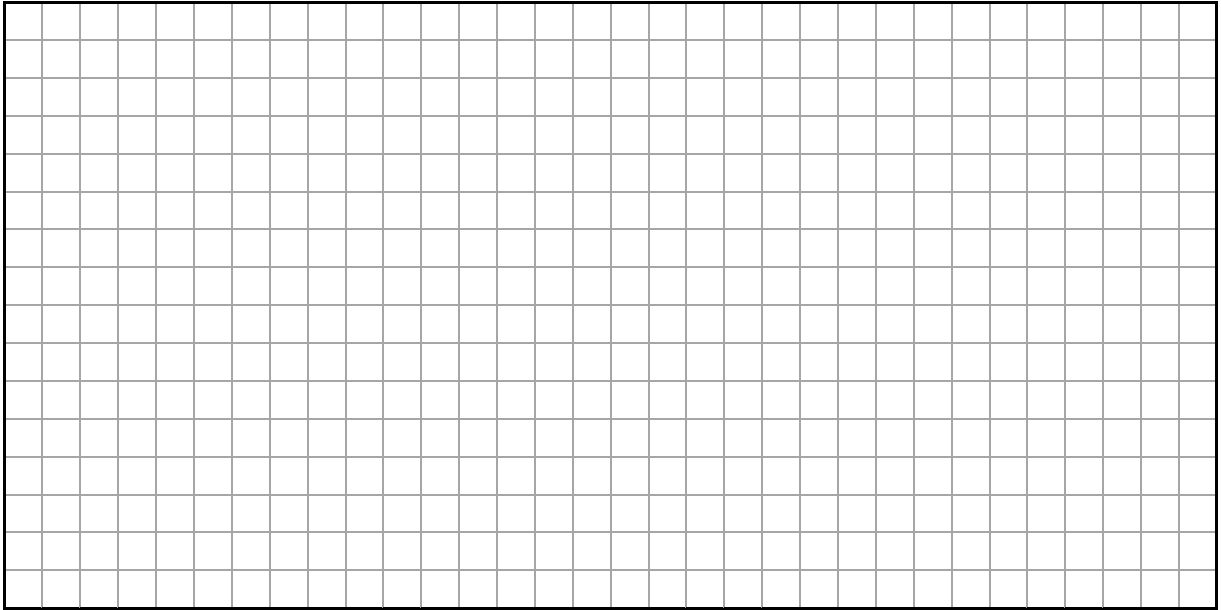
- (a) Construct a triangle ABC , where $|AB| = 8$ cm and $|AC| = |BC| = 6$ cm.
Show all of your construction lines clearly.



- (b) Construct the **perpendicular bisector** of the side AB that passes through the point C .
- (c) Use the **Theorem of Pythagoras** to work out the perpendicular distance from C to AB .
Give your answer in the form $a\sqrt{b}$ cm, where $a, b \in \mathbb{N}$.



- (d) Use **trigonometry** to work out $|\angle CAB|$. Give your answer correct to the nearest degree.



Question 11 (Suggested maximum time: 10 minutes)

Question 11 (Suggested maximum time: 10 minutes)

A group of 216 music students were asked what musical instrument was their favourite to play.

25% of the students said the guitar was their favourite instrument.

$\frac{3}{8}$ of the students said the violin was their favourite instrument.

$\frac{1}{3}$ of the students said their favourite instrument was the piano.

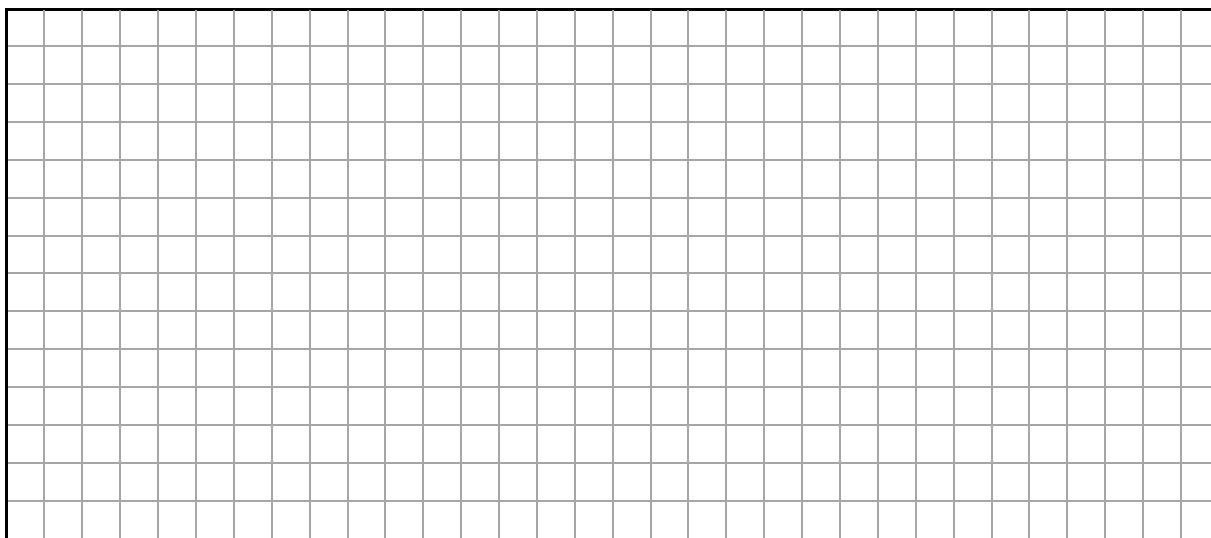
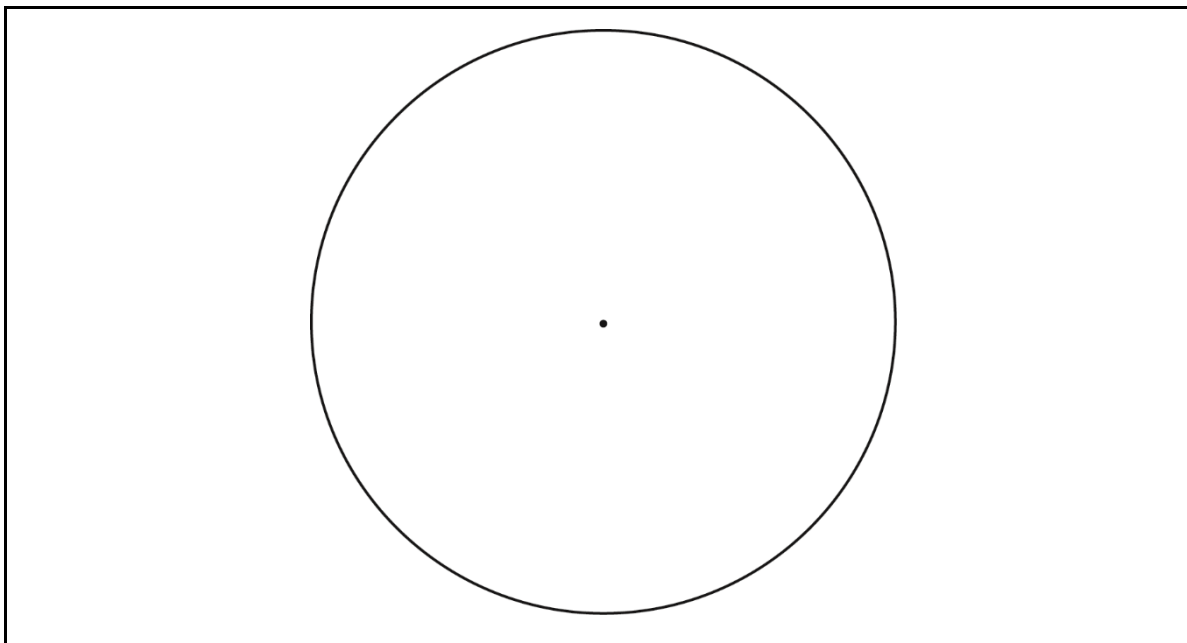
The remainder of the students in the group said that the flute was their favourite instrument.

- (a) Complete the frequency table to show the number of music students who chose each instrument as their favourite.

Instrument	Guitar	Violin	Piano	Flute
Frequency				

[illegible]

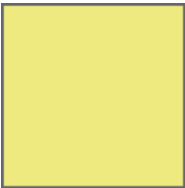
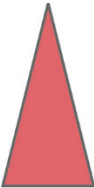
- (b) Complete the **pie chart** below to summarise the results, showing the proportion of music students who chose each instrument as their favourite.
Label each sector **and** the size of each angle clearly.
Show any working out and construction lines.



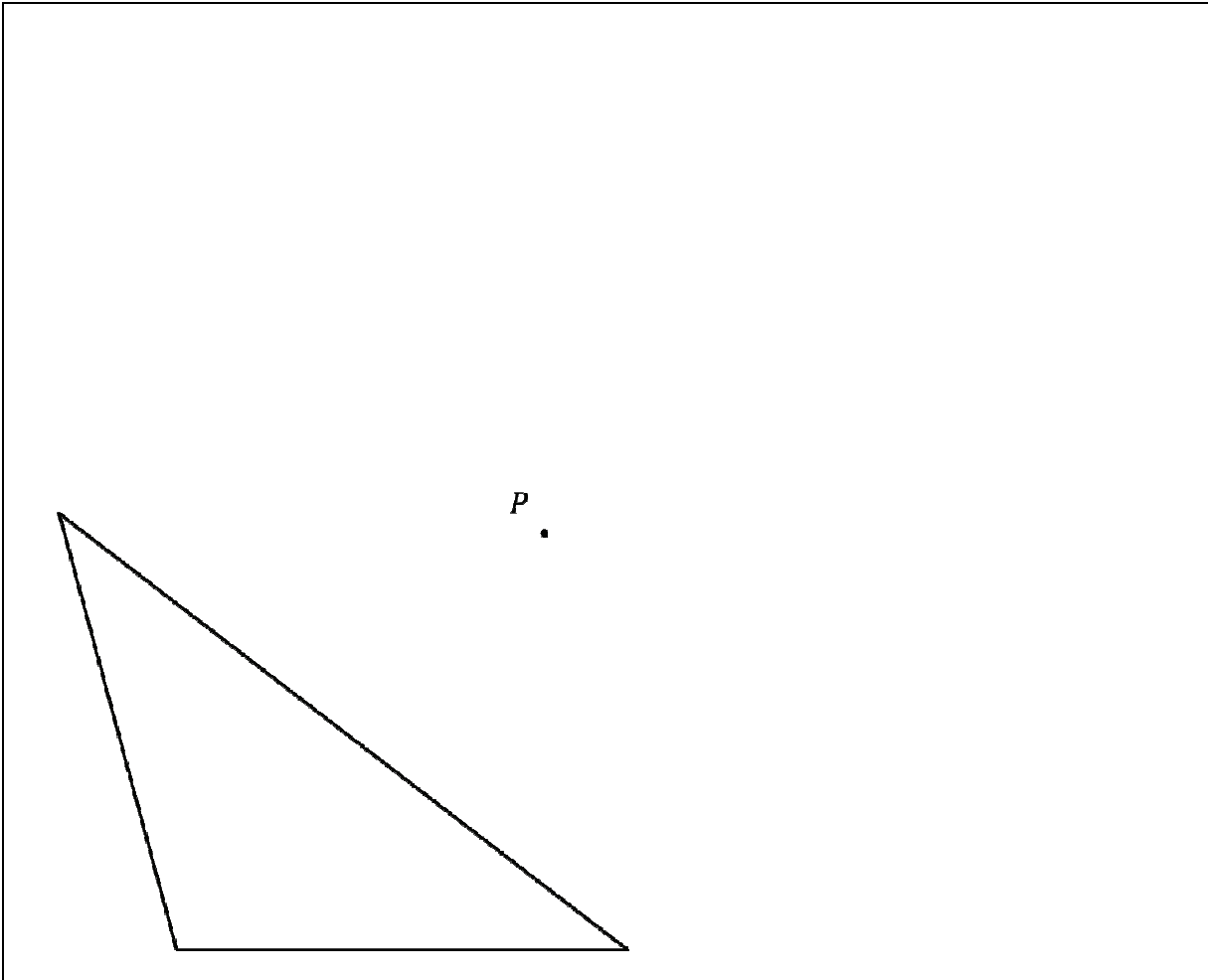
Question 12

(Suggested maximum time: 5 minutes)

- (a) How many **axes of symmetry** does each of the following shapes have?
Fill your answers into the table below.

Shape				
Name	Square	Isosceles Triangle	Regular Pentagon	Rectangle
Number of axes of symmetry				

- (b) Draw the image of the object under S_P , **central symmetry** through the point P .



Page for extra work.

Label any extra work clearly with the question number and part.

[illegible]

Label any extra work clearly with the question number and part.

[illegible]

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Time: 2 hours



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